

Quadrature Rotary Encoder

Make every Gray-code edge and invalid jump visible

Count contact transitions, not vendor-dependent clicks

Lesson	032 — component	Time	120–180 minutes
Prerequisites	Lessons 001, 002, 006, 016	Board	Arduino Mega 2560
Evidence	TP-A, TP-B, position nibble, RGB state	Status	host verified; bench open
Energy class	E1: USB, passive contacts, resistor-limited LEDs		

Specimen gate. The Elegoo listing authorizes the rotary encoder product family, not a universal module pin order or electrical revision. Identify the exact markings and pins before wiring. Do not infer CLK, DT, switch, supply, or ground order from a photograph or retail family name.

1 Mission and learning outcomes

phase A + phase B → Gray-code edge → delta + position + diagnostics.

The unit is one valid edge, not a mechanical detent or “click.” By the end, you can:

- trace both valid Gray-code directions from any starting phase;
- distinguish repeated states, bounce reversals, and invalid jumps;
- explain why the decoder establishes a baseline without movement;
- verify transactional acquisition and reverse-order shutdown;
- separate electrical phase evidence from software presentation;
- record host evidence without claiming a physical result.

2 Literal wiring and unpowered inspection

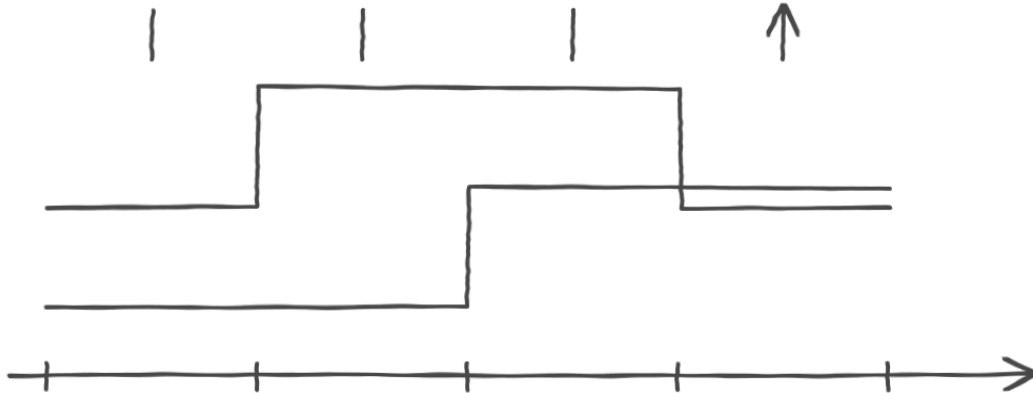
Role	Mega pin	Literal path
phase A / TP-A	D24	identified CLK/A contact to D24; internal pull-up
phase B / TP-B	D25	identified DT/B contact to D25; internal pull-up
separate switch	D26	identified SW contact to D26; separate Button
position nibble	D30–D33	pin to 1 kΩ, LED anode; cathode to GND
RGB decision state	D5–D7	each channel through its own 1 kΩ resistor
reference	GND	identified module contact ground to Mega GND

The encoder component owns only D24 and D25. D26 remains compositionally separate so the same decoder works with encoders that have no push switch. The four nibble LEDs show the low position bits. The RGB LED pulses green for a valid edge, remains amber for no edge, and pulses red for an invalid jump. Cadence and the nibble make the evidence interpretable without color alone.

Unpowered prediction. Record the exact module markings and revision.

Identify ground: _____. Verify continuity from D24 and D25 to the intended contacts, the separate D26 switch path, common ground, polarity, and one resistor per LED channel.

3 Gray-code timing drawing



Upper trace: **A**. Lower trace: **B**. Samples: **00** → **01** → **11** → **10** → **00**. Four valid clockwise edges.

Alternative text: phase B rises, phase A rises, phase B falls, then phase A falls, producing the sampled masks 00, 01, 11, 10, 00 and four clockwise edge decisions. Reverse traversal reports counterclockwise.

Before power, predict the TP-A and TP-B levels for one slow traversal.

4 Decoder contract

Observation	Deterministic result
valid clockwise edge	delta = +1; increment position
valid counterclockwise edge	delta = -1; decrement position
same phase	zero delta; preserve position and invalid count
two-bit jump	zero delta; increment invalid count; adopt new baseline
position limit	hold at INT32_MIN/MAX; report saturation

reversed exchanges the reported signs. Bounce reversals are decoded literally; the component does not guess a detent. The invalid-transition count saturates at 65,535. Position saturates at its signed 32-bit limits and never wraps. **resetPosition()** changes only accumulated position and clears position saturation; it preserves the current hardware baseline.

Initialization validates distinct Mega digital pins before claims, acquires A then B, configures the selected pull, and reads A then B exactly once to set a movement-free baseline. Each update again reads A once and then B once.

5 Lifecycle and failure evidence

If B acquisition or configuration fails, initialization rolls A back. Repeated successful initialization performs no new hardware work. Shutdown releases B then A, is idempotent, and retains count evidence while the

snapshot status becomes `NotInitialized`. Destruction follows the same cleanup. Copy and move operations are prohibited.

Evidence question	Required proof
Did ownership succeed?	ready indication only after both phase inputs initialize
Did decoding succeed?	TP sequence agrees with delta and position nibble
Was a jump rejected?	unchanged position plus incremented count and red pulse
Did shutdown become safe?	RGB/nibble inactive and both input modes released

An inactive LED does not prove input release. Record acquisition and post-shutdown input-mode evidence separately.

6 Narrative Mega example

```
observeEncoder();
decideEncoderEvidence();
presentEncoderEvidence();
```

Setup acquires phase inputs, the separate button, nibble LEDs, and RGB indicator; configures every output inactive; initializes the decoder; then shows ready. The loop observes exactly one phase pair, decides from one stable snapshot, and presents its low nibble and diagnostic state.

7 Predict, observe, interpret

1. Predict the complete TP-A/TP-B sequence before applying USB power.
2. Rotate slowly through one full electrical sequence. Observe TP-A and TP-B, then record four signed edges independent of mechanical detents.
3. Compare the electrical sequence, low position nibble, and valid-edge pulse. Interpret agreement as decoder evidence.
4. Reverse direction and deliberately include a small bounce reversal. Predict literal opposing edges rather than hidden filtering.
5. Use only an unpowered, pre-inspected contact fixture to arrange a safe two-bit jump. Predict unchanged position, one invalid increment, and a red pulse after power is restored.
6. Invoke shutdown separately. Record inactive displays and input release, then remove USB power.

8 Diagnosis and stop conditions

Observation	Action and interpretation
count direction reversed	verify A/B identity; use configuration reversal only intentionally
two bits change together	inspect contacts and sampling; expect invalid evidence
LED count disagrees with TP sequence	stop and inspect mapping before proceeding
position changes on an invalid jump	software evidence has failed; do not accept
hot part, resets, unstable or out-of-rail TP	remove USB immediately
unidentified module contact	remain unpowered until primary evidence identifies it

Never move jumpers while powered. Software shutdown is lifecycle behavior, not an emergency stop; remove USB power for any unsafe or unexplained condition.

9 Open hardware acceptance record

Board/revision	_____	Commit	_____
Encoder markings	_____	Pin source	_____
Supply/instrument	_____	Tool versions	_____
Operator/date	_____	Deviations	_____

- ☐ unpowered identity, continuity, polarity, and resistor check;
- ☐ acquisition prediction, observation, and interpretation;
- ☐ both direction sequences and all four starting phases;
- ☐ repeated state, bounce reversal, invalid jump, and recovery;
- ☐ TP evidence and LED presentation compared independently;
- ☐ shutdown input release, reset, and power removal recorded.

Prediction: _____

Observation: _____

Interpretation and limits: _____

10 Exercises

- Starting at 11, list the next masks and deltas for one reverse cycle.
- Explain why adopting the new phase after an invalid jump enables recovery without inventing movement.
- Predict the nibble at positions -1 , 0 , 7 , 15 , and 16 .
- Design a trace that distinguishes a repeated sample from contact bounce.
- Explain why the separate switch belongs to `Button`, not the quadrature decoder.

11 Reproduce the software evidence

```
make host-test
make host-test-exceptions
make arduino-Lesson032QuadratureEncoder
make size-check
make lessons
make lessons-check
make site-check
make hardware-card LESSON=032 BOARD=mega2560
```

Tests cover all starting phases, directions, reversal, partial turns, bounce, repeated states, every invalid jump and recovery, both saturations, reset, exact read order, transactional failures, lifecycle, traits, and replay. Compilation and deterministic tests are software evidence, not a physical module acceptance result.

12 Reference trail

- ADK `quadrature_encoder.h`, focused tests, and canonical Lesson 032 sketch;

- ADK Lessons 031–033 design record and authorized Elegoo family ledger;
- Arduino Mega 2560 Rev3 datasheet for the exact board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and a monochrome timing drawing but does not claim PDF/UA conformance.